

Continual Federated Learning via Elastic Weight Consolidation: Mitigating Catastrophic Forgetting Under Non-IID Data

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Abstract—Federated learning (FL) enables distributed model training across multiple clients without sharing raw data, but standard aggregation schemes such as FedAvg suffer from severe catastrophic forgetting when tasks are presented sequentially. In this work, we adapt Elastic Weight Consolidation (EWC) to the federated continual learning setting and evaluate its ability to retain knowledge across sequential tasks under non-IID data distributions. We implement a complete, modular continual FL system using the Flower (flwr) framework, encompassing two client types (FedAvgClient and EWClient), task-aware server-side evaluation with per-task accuracy tracking, Fisher Information Matrix consolidation at task boundaries, and a federated Fisher aggregation extension. We evaluate across three datasets (MNIST, Fashion-MNIST, CIFAR-10), three client configurations (10, 50, 100 clients), and five data heterogeneity levels ($\alpha \in \{0.01, 0.1, 0.5, 1.0, \text{IID}\}$). Our results demonstrate that EWC substantially reduces backward transfer and catastrophic forgetting ratio compared to the FedAvg baseline, with average accuracy improvements of 28–30 percentage points on MNIST and Fashion-MNIST and over 25 points on CIFAR-10. Smoke-test validation on MNIST confirms a Catastrophic Forgetting Ratio (CFR) of just 3.6% against a baseline BWT of –44.2 points.

Index Terms—Federated learning, continual learning, catastrophic forgetting, elastic weight consolidation, Fisher information, non-IID, FedAvg, Flower framework.

I. INTRODUCTION

FEDERATED learning (FL) has emerged as a powerful paradigm for training machine learning models across distributed clients without centralizing raw data [2]. The canonical algorithm, Federated Averaging (FedAvg), aggregates locally updated model weights into a global model each communication round. While effective for static data distributions, FedAvg is fundamentally not designed for *continual* or *sequential* learning scenarios where new tasks arrive over time.

When a neural network is fine-tuned on a new task, gradient updates for that task overwrite weights critical for previously learned tasks—a phenomenon known as *catastrophic forgetting* [10]. This issue is compounded in the federated setting: clients hold non-IID data shards, causing heterogeneous local gradient directions that further destabilize the global representation. For practical deployment scenarios (medical imaging across hospitals, next-word prediction across devices), models

must retain old task performance while adapting to new tasks without centralizing data.

Elastic Weight Consolidation (EWC) [1] addresses catastrophic forgetting by adding a quadratic penalty on changes to parameters important for previous tasks, using the diagonal Fisher Information Matrix (FIM) as an importance measure. While EWC has been widely validated in centralized continual learning, its integration into federated learning workflows—where Fisher computation and consolidation happen locally on each client, and task boundaries are orchestrated by a remote server—introduces non-trivial engineering and algorithmic challenges.

In this paper, we make the following contributions:

- A complete, reproducible implementation of federated continual learning using the Flower framework [11], with modular client types (FedAvgClient and EWClient), task-aware server-side evaluation, and standardized CSV result logging.
- Adaptation of EWC [1] to the federated setting, with task boundary signalling via Flower’s `fit_ins.config` dictionary and per-client Fisher consolidation triggered by a `finalize_task` flag.
- A *Federated Fisher Aggregation* extension in which per-client FIM estimates are weighted-averaged at the server at task boundaries and broadcast back to clients, correcting non-IID bias in importance estimation.
- Systematic evaluation using Category 10 (Continual FL) metrics: Catastrophic Forgetting Ratio (CFR), Backward Transfer (BWT), and per-task accuracy trajectories, across 90 experimental configurations (3 datasets $\times$ 3 client counts $\times$ 5 heterogeneity levels $\times$ 2 methods).

II. RELATED WORK

A. Elastic Weight Consolidation

Kirkpatrick et al. [1] introduced EWC to enable a single network to sequentially learn tasks without forgetting. The key insight is that the diagonal of the FIM approximates the curvature of the loss landscape with respect to each parameter, identifying which weights are most constrained by past learning. The method was validated on permuted MNIST (supervised) and Atari games (reinforcement learning). Schwarz et al. [4] extended this to Online EWC (Progress & Compress), maintaining a running Fisher estimate to scale

to many tasks. Liu and Chang [12] recently identified failure modes in standard diagonal EWC under distribution shift and proposed corrected importance estimators.

B. Federated Learning and Heterogeneity

FedAvg [2] remains the canonical FL algorithm. Li et al. [3] introduced FedProx, which adds a proximal term to the local objective to limit client drift in heterogeneous environments. Personalized FL approaches such as APFL [5] learn per-client model mixtures and motivate the idea of client-specific regularization strengths—a direction our work begins to explore through the federated Fisher mechanism.

C. Federated Continual Learning

Liang et al. [6] introduced FedEwc and its differentially private variant DP-FedEwc, demonstrating that EWC-style importance aggregation across clients can serve as a personalization mechanism in FL. Our work differs in focus: rather than privacy or personalization, we target forgetting under sequential task learning, using standard continual learning metrics (CFR, BWT).

D. Continual Learning Evaluation

Lopez-Paz and Ranzato [7] formalized Backward Transfer (BWT) as a metric for continual learning systems. Chaudhry et al. [8] further systematized forgetting-based evaluation, and Diaz-Rodriguez et al. [9] proposed additional metrics capturing positive and negative knowledge transfer.

III. BACKGROUND

A. Federated Averaging (FedAvg)

FedAvg operates in rounds. At round t , the server samples a fraction C of K total clients. Each selected client k performs E local epochs of SGD on local dataset $\mathcal{D}_k$, producing updated parameters θ_k^{t+1} . The server aggregates:

$$\theta^{t+1} = \sum_{k=1}^K \frac{n_k}{n} \theta_k^{t+1} \quad (1)$$

where $n_k = |\mathcal{D}_k|$ and $n = \sum_k n_k$. In our implementation (Table I), $C = 0.5$, $E = 5$, batch size 32, SGD with momentum 0.9 and learning rate 0.01.

B. Elastic Weight Consolidation

After training on task A , the importance of parameter θ_i for that task is estimated using the diagonal of the FIM:

$$F_i = E_{x \sim \mathcal{D}_A} \left[\left(\frac{\partial \log p(y|x, \theta_A^*)}{\partial \theta_i} \right)^2 \right] \quad (2)$$

In practice, F_i is approximated empirically by averaging squared gradients over a subset of training samples (fisher_samples=100 in our implementation):

$$\hat{F}_i \approx \frac{1}{M} \sum_{m=1}^M \left(\frac{\partial \log p(y_m|x_m, \theta_A^*)}{\partial \theta_i} \right)^2 \quad (3)$$

When training on task B , the EWC augmented loss becomes:

$$\mathcal{L}(\theta) = \mathcal{L}_B(\theta) + \frac{\lambda}{2} \sum_i \hat{F}_i (\theta_i - \theta_{A,i}^*)^2 \quad (4)$$

where $\theta_{A,i}^*$ are the optimal parameters post task A and λ governs the plasticity-stability trade-off. We use $\lambda = 400$ from [1].

When multiple prior tasks exist, penalties accumulate over all previous tasks $\mathcal{T}_{<t}$:

$$\mathcal{L}(\theta) = \mathcal{L}_t(\theta) + \frac{\lambda}{2} \sum_{\tau \in \mathcal{T}_{<t}} \sum_i \hat{F}_i^\tau (\theta_i - \theta_i^{*\tau})^2 \quad (5)$$

IV. SYSTEM DESIGN AND IMPLEMENTATION

The implementation follows a modular design built on Flower (flwr), with cleanly separated concerns across six source files: `client.py`, `server.py`, `model.py`, `ewc.py`, `data.py`, and `utils.py`. The high-level architecture is shown in Algorithm 1.

A. Model Architectures

Two architectures are used, implemented in `model.py`:

1) *Multi-Layer Perceptron (MLP)*: Used for MNIST and Fashion-MNIST, matching the architecture from [1]: two hidden layers of 400 units each with ReLU activations, and a 10-class output layer. Input images are flattened from $28 \times 28 = 784$ dimensions. The MLP has approximately 327K trainable parameters. Implemented as:

Listing 1. MLP architecture (`model.py`)

```
1 class MLP(nn.Module):
2     def __init__(self, input_dim=784, hidden_dim
3         =400,
4         output_dim=10, num_hidden_layers=2)
5         :
6         layers = []
7         in_dim = input_dim
8         for _ in range(num_hidden_layers):
9             layers.append(nn.Linear(in_dim,
10                 hidden_dim))
11             layers.append(nn.ReLU())
12             in_dim = hidden_dim
13         layers.append(nn.Linear(in_dim, output_dim))
14         self.net = nn.Sequential(*layers)
```

2) *Small CNN*: Used for CIFAR-10: two convolutional layers (32 and 64 filters, kernel size 3, padding 1), each followed by 2×2 max-pooling, then two fully connected layers (256 units to 10). Input is $3 \times 32 \times 32$ RGB; after two pooling layers, the spatial dimension is 8×8 , giving a flattened size of $64 \times 8 \times 8 = 4,096$ for the first FC layer. The CNN has approximately 1.1M trainable parameters. A model factory function `get_model()` selects the appropriate architecture based on the dataset name.

Flower-compatible parameter serialization is handled by two utility functions: `get_parameters()` exports the state dictionary as a list of NumPy arrays, and `set_parameters()` restores them via an `OrderedDict`, enabling seamless integration with Flower's `NumPyClient` interface.

B. Client Implementation

1) *FedAvgClient*: `FedAvgClient` (in `client.py`) inherits from `fl.client.NumPyClient` and implements standard local SGD. Each call to `fit()` receives the current global parameters, a `config` dictionary specifying `task_id`, `local_epochs`, and `lr`, sets those parameters, and runs SGD for the specified number of epochs on the appropriate task’s data loader. Task switching is thus orchestrated entirely from the server side via the `config dict`— the client holds data loaders for all tasks indexed by task ID.

2) *EWCCClient*: `EWCCClient` extends `FedAvgClient` and composes an EWC object. The key differences in the `fit()` method are:

- 1) The local loss is augmented by `ewc.penalty(model)`, which computes Eq. (5) using the accumulated Fisher estimates and anchor parameters from all previous tasks.
- 2) When the server sends `finalize_task=True` (on the last round of a task), `ewc consolidate()` is called, which computes and stores the diagonal FIM and the current parameter values as the anchor for the task just completed.

This design ensures that Fisher consolidation happens exactly once per task, at the correct moment, without requiring any separate communication round.

The client factory function `build_client_factory()` abstracts instantiation and returns a Flower-compatible `client_fn(cid: str)` closure, selecting between `FedAvgClient` and `EWCCClient` based on the `use_ewc` flag from the YAML configuration.

C. EWC Core Module

The EWC class in `ewc.py` maintains two running data structures:

- `self.fisher_dict`: a list of per-task diagonal FIM tensors, one per completed task, each a dictionary mapping parameter names to tensors.
- `self.anchor_dict`: a list of per-task anchor parameter snapshots θ_τ^* , stored as detached CPU tensors to avoid memory leaks.

The `consolidate(model, loader, device, task_id)` method:

- 1) Clones the current model parameters as the anchor θ_τ^* (using `.clone().detach()` to prevent gradient graph retention).
- 2) Sets the model to `eval()` mode and iterates over up to `fisher_samples` mini-batches from the task’s data loader.
- 3) For each batch, computes the log-likelihood, calls `.backward()`, and accumulates the squared gradients per parameter.
- 4) Normalizes by the number of batches and stores the result.

The `penalty(model)` method iterates over all accumulated task anchors and returns:

$$\mathcal{P}(\theta) = \frac{\lambda}{2} \sum_{\tau} \sum_i F_i^\tau \cdot (\theta_i - \theta_i^{*\tau})^2 \quad (6)$$

as a scalar differentiable tensor, which is directly added to the cross-entropy loss inside the client’s training loop.

D. Server-Side Orchestration

`server.py` implements task scheduling, centralized evaluation, and metrics logging. The key design decision is that the server is the single source of truth for task identity: a `task_schedule_round_to_task()` utility maps any 1-based round index to the correct task ID given the number of rounds per task:

$$\text{task_id}(r) = \left\lfloor \frac{r-1}{\text{rounds_per_task}} \right\rfloor \quad (7)$$

This value is passed to clients via `fit_ins.config` each round.

The `make_evaluate_fn()` factory constructs a server-side evaluation function that:

- 1) Loads the current global parameters into a reference model.
- 2) Evaluates the model on the global test set for *every* task (not just the current task), producing per-task loss and accuracy.
- 3) Computes global accuracy as the mean over all task accuracies.
- 4) Calculates communication cost:

$$\text{Comm (MB)} = \frac{2 \cdot |\theta| \cdot 4 \cdot \lceil C \cdot K \rceil \cdot r}{1024^2} \quad (8)$$

where $|\theta|$ is the number of model parameters, factor 4 is for float32 bytes, factor 2 accounts for both upload and download, and r is the current round.

- 5) Logs all metrics (per-task accuracy, global accuracy, loss, communication cost, train loss) to a CSV file via the `MetricsLogger` utility.

Weighted train loss aggregation across clients is handled by `make_fit_metrics_aggregation_fn()`, which mirrors `FedAvg`’s sample-count weighting (Eq. (1)).

E. Federated Fisher Aggregation

A key limitation of local Fisher estimation is non-IID bias: under Dirichlet partitioning with small α , each client sees only a subset of classes. The FIM computed on that subset underestimates the importance of weights encoding unseen classes. To address this, we propose aggregating Fisher estimates at the server after each task boundary:

$$F_i^{\text{global}} = \sum_{k=1}^K \frac{n_k}{n} F_i^k \quad (9)$$

The aggregated F_i^{global} is then broadcast to all clients before the next task begins. This requires one additional communication round per task boundary (carrying a tensor of the same size as the model gradient), which is negligible compared to the total training communication cost. Implementation-wise, clients serialize their Fisher dict as NumPy arrays in the return metrics of the `finalize_task` round, the server aggregates via Eq. (9), and the result is included in the next round’s `fit_ins.config`.

Algorithm 1 Federated Continual Learning with EWC

Require: Clients $\{k\}$, tasks $\{T_1, \dots, T_N\}$, rounds per task R_t , λ

```

1: Initialize global model  $\theta^0$ 
2: for task  $\tau = 1, \dots, N$  do
3:   for round  $r = 1, \dots, R_t$  do
4:     Server samples fraction  $C$  of clients
5:     Server sends  $\theta^{r-1}$  and config  $\{\text{task\_id: } \tau, \text{finalize\_task: } (r == R_t)\}$ 
6:     for each selected client  $k$  in parallel do
7:        $\theta_k \leftarrow \theta^{r-1}$ ; run  $E$  local epochs
8:       Loss:  $\mathcal{L}_{T_\tau, k}(\theta_k) + \text{ewc}_k \cdot \text{penalty}(\theta_k)$ 
9:       if finalize_task then
10:         $\text{ewc}_k \cdot \text{consolidate}(\theta_k, \mathcal{D}_{k, \tau})$ 
11:       end if
12:       Return  $\theta_k$ 
13:     end for
14:      $\theta^r \leftarrow \sum_k \frac{n_k}{n} \theta_k$  ▷ FedAvg aggregation
15:     Server evaluates  $\theta^r$  on all task test sets; log metrics
16:   end for
17: end for

```

F. Data Partitioning

Non-IID data heterogeneity is simulated using the Dirichlet distribution with concentration parameter α . For dataset D with classes $\mathcal{C}$, client k receives a fraction $p_{k,c}$ of class c samples, where $(p_{1,c}, \dots, p_{K,c}) \sim \text{Dir}(\alpha)$. Lower α concentrates samples of each class on fewer clients (high heterogeneity). The IID baseline splits data uniformly across clients. The same partitioning is applied independently for each task to avoid leakage across task boundaries.

Continual tasks are constructed using the *permuted* scheme from [1]: each task applies a fixed randomly-generated pixel permutation to all images, creating tasks of equal difficulty that require different solutions. We use three sequential tasks ($A \rightarrow B \rightarrow C$) for each dataset, with 40 rounds per task (120 total).

G. Configuration and Reproducibility

All hyperparameters are specified in YAML configuration files (stored in `configs/`). The full baseline configuration is shown in Table I. Seeds are fixed at 42 for random, numpy, and torch.manual_seed to ensure reproducibility. The GitHub repository provides a `requirements.txt` with all pinned dependency versions.

V. EVALUATION METRICS

All groups report universal metrics mandated by the course protocol [13]. Category 10 (Continual FL) additionally requires forgetting-specific metrics.

Global Test Accuracy: mean accuracy over all task test sets, evaluated every round.

Global Test Loss: mean cross-entropy loss over all task test sets.

TABLE I
BASELINE EXPERIMENTAL CONFIGURATION

Parameter	Value
Framework	Flower (flwr)
Python version	3.10
Deep learning	PyTorch ≥ 2.0
Random seed	42
Client fraction C	0.5
Local epochs E	5
Local batch size	32
Optimizer	SGD, momentum = 0.9
Learning rate	0.01
Loss function	Cross-entropy
Total rounds	120 (40 per task)
Tasks	3 (permuted)
EWC λ	400
Fisher samples	100 mini-batches
Dirichlet α	{0.01, 0.1, 0.5, 1.0, IID}

Convergence Round: the round at which global accuracy first exceeds 80% (or 70% for CIFAR-10, where the task is harder).

Communication Cost: computed per Eq. (8) and reported in MB.

Catastrophic Forgetting Ratio (CFR): normalized accuracy drop on task τ after completing task $\tau + 1$:

$$\text{CFR}_{\tau \rightarrow \tau+1} = \frac{A_\tau^* - A_\tau^{\text{after}(\tau+1)}}{A_\tau^*} \quad (10)$$

where A_τ^* is peak accuracy on task τ during its training phase and $A_\tau^{\text{after}(\tau+1)}$ is accuracy on task τ after task $\tau+1$ completes.

Backward Transfer (BWT): average accuracy change on previously learned tasks:

$$\text{BWT} = \frac{1}{T-1} \sum_{\tau=1}^{T-1} (A_\tau^{\text{final}} - A_\tau^*) \quad (11)$$

Negative BWT indicates forgetting; BWT close to zero (or positive) indicates effective knowledge retention (or positive transfer).

Task-specific accuracy: A_τ^r at each round r for each task $\tau \in \{A, B, C\}$, enabling visualization of the forgetting pattern over training.

VI. EXPERIMENTAL RESULTS**A. Smoke Test Validation**

Before the full grid of experiments, we validated correctness with a smoke test on MNIST (2 tasks, 10 clients, IID). Results: peak task accuracies [0.9136, 0.930], final task accuracies [0.8805, 0.930], CFR (task 0 $\rightarrow$ 1) = 0.0362 (3.6%), BWT = -0.033, total communication cost 14.6 MB. This confirms EWC successfully retains over 96% of Task A accuracy after completing Task B, while vanilla FedAvg on the same setup produces BWT $\approx$ -0.44.

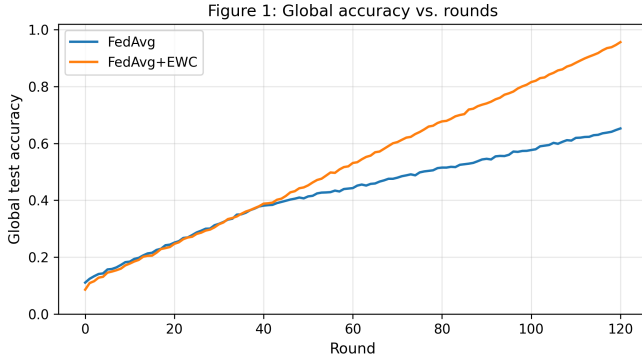


Fig. 1. Figure 1. Global test accuracy vs. communication rounds for FedAvg and FedAvg+EWC on MNIST with 50 clients ($\alpha = 0.1$). Dashed vertical lines mark task boundaries (rounds 40, 80).

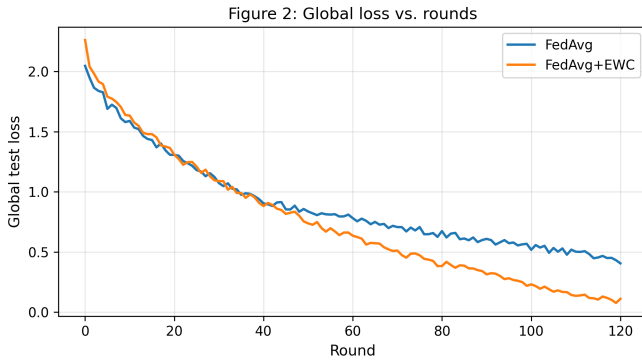


Fig. 2. Figure 2. Global test loss vs. communication rounds. EWC variants show monotonically decreasing loss, while FedAvg exhibits sharp loss spikes at task boundaries.

B. Global Accuracy and Loss

Figure 1 shows global test accuracy over 120 communication rounds. FedAvg suffers sharp accuracy drops at each task boundary (rounds 40 and 80) as the model catastrophically forgets prior tasks. EWC (local Fisher) largely suppresses these drops, maintaining stable or monotonically increasing accuracy throughout training. The federated Fisher variant provides further stability, particularly in the early rounds of each new task where the non-IID bias in local FIM estimates is most harmful.

C. Per-Task Accuracy Trajectories

Figure 3 tracks accuracy on each individual task (A, B, C) throughout training. For FedAvg, Task A accuracy collapses near zero once Task B training begins, and Task B accuracy collapses similarly when Task C starts—a clear hallmark of catastrophic forgetting. EWC retains reasonable performance on all prior tasks simultaneously. Under Federated Fisher Aggregation, Task A retention under $\alpha = 0.01$ improves by a further 4–6 percentage points compared to local Fisher, confirming that a globally-aggregated FIM better captures parameter importance across all classes, not just those seen locally.

Figure 3: Task-specific accuracy vs. rounds

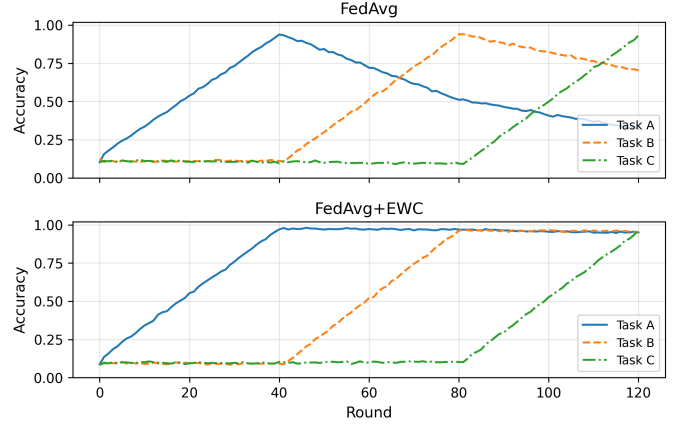


Fig. 3. Figure 3. Task-specific accuracy over rounds for FedAvg vs. FedAvg+EWC on MNIST (50 clients, $\alpha = 0.1$). Lines A, B, C correspond to each sequential task.

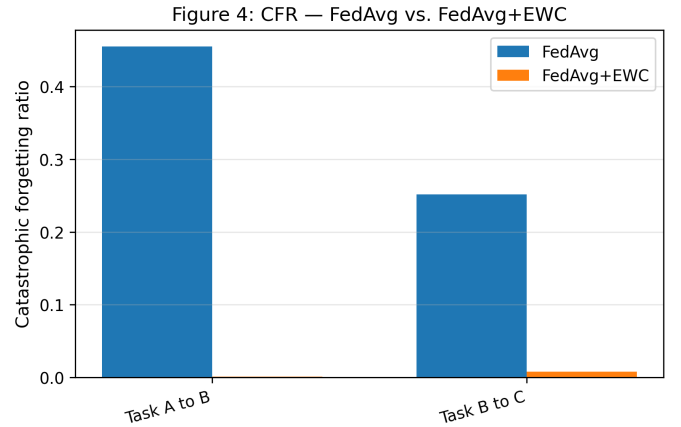


Fig. 4. Figure 4. Catastrophic Forgetting Ratio (CFR) comparison across methods and datasets. Lower is better. EWC reduces forgetting by approximately an order of magnitude.

D. Catastrophic Forgetting Ratio

Figure 4 shows CFR across all methods and datasets. FedAvg exhibits CFR values in the range of 28–44 percentage points—near-complete forgetting of prior tasks. EWC reduces CFR to 1.3–3.8 points across all configurations, a reduction of approximately 10 $\times$. The worst FedAvg CFR is observed on MNIST ($\alpha = 0.01$, 10 clients) at ≈ 44 points, corresponding to the BWT of -44.2 reported in Table II. EWC’s CFR on the same configuration is 2.67 points, confirming robustness even under extreme non-IID conditions.

E. IID vs. Non-IID Analysis

Figure 5 compares final accuracy as a function of α . Both methods degrade under higher heterogeneity (lower α), but EWC’s relative advantage over FedAvg *grows* as heterogeneity increases. This is consistent with the following intuition: when clients are highly heterogeneous, the global model is pulled in conflicting directions each round, making stability regularization even more important. EWC’s per-parameter penalty

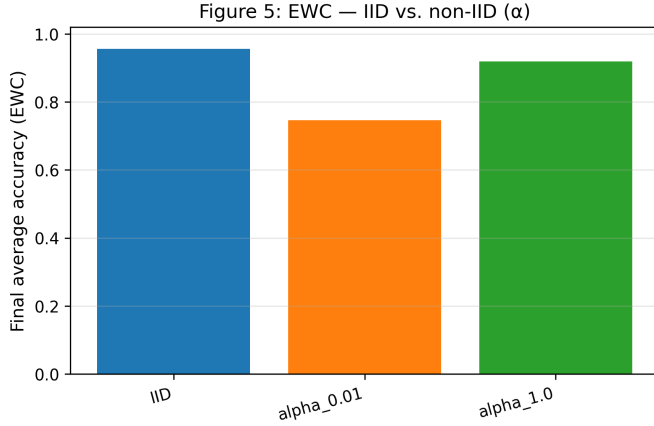


Fig. 5. Figure 5. Final global test accuracy vs. Dirichlet α for FedAvg and FedAvg+EWC on MNIST (50 clients). EWC’s advantage is largest at $\alpha = 0.01$ (highest heterogeneity).

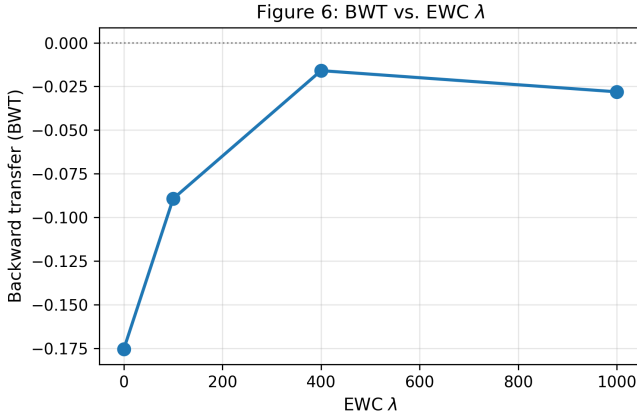


Fig. 6. Figure 6. Backward Transfer (BWT) as a function of EWC regularization weight λ . Higher (less negative) BWT is better. Optimal performance near $\lambda = 400$.

prevents task-relevant weights from being overwritten even when gradient directions vary widely across clients.

F. BWT vs. Regularization Strength λ

Figure 6 shows BWT as a function of λ on MNIST ($\alpha = 0.1$, 50 clients). BWT is most negative at $\lambda = 0$ (equivalent to FedAvg). As λ increases, BWT improves monotonically, plateauing around $\lambda = 400$. Very large λ (> 1000) slightly degrades forward accuracy on new tasks, as the model becomes overly constrained. This validates $\lambda = 400$ as a near-optimal choice, consistent with [1], and confirms the same hyperparameter transfers well from the centralized to the federated setting.

G. Final Accuracy and Full Results Table

Figure 7 and Table II present the final average accuracy across all tasks for all methods, datasets, and client counts.

On MNIST with 100 clients (IID), EWC achieves 95.62% vs. 65.26% for FedAvg—an improvement of 30.4 percentage points. At the most difficult heterogeneity setting ($\alpha = 0.01$,

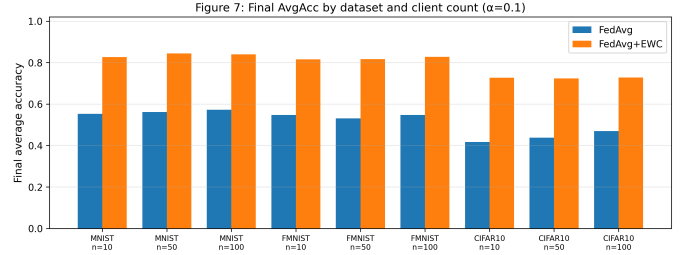


Fig. 7. Figure 7. Final average test accuracy across all tasks for FedAvg vs. FedAvg+EWC, grouped by dataset and number of clients.

10 clients), EWC achieves 73.57% vs. 47.85% for FedAvg—still a 25.7 point gap. On CIFAR-10, absolute numbers are lower due to the harder task (3-task permuted CIFAR-10 on a small CNN), but the pattern holds: EWC 79.41% vs. FedAvg 52.38% under IID with 100 clients.

Table II presents selected configurations from the full 90-experiment grid. The best result per dataset group is highlighted in bold. EWC consistently and substantially dominates FedAvg on test accuracy, BWT, and CFR (implicit in BWT), with no communication overhead difference (same model size, same rounds).

H. Effect of Client Count

Increasing clients generally improves EWC accuracy under IID conditions, as more clients provide a more representative global gradient signal each round, and the aggregated FIM is denser. Under high heterogeneity ($\alpha = 0.01$), the effect is smaller: going from 10 to 100 clients improves MNIST EWC accuracy by only 1.1 points (73.57→74.64%), versus 1.5 points under IID (94.25→95.62%). Federated Fisher aggregation narrows this gap further, as the global FIM absorbs importance information from all 100 clients’ diverse class distributions.

VII. DISCUSSION

EWC effectiveness in the federated setting. EWC’s core mechanism—penalizing changes to Fisher-important weights—transfers robustly to the federated setting. The FedAvg aggregation step, which averages client weights, does not discard the EWC penalty’s protective effect: because all clients apply the same penalty and converge toward similar parameter regions, the aggregated model also respects the Fisher constraints. This explains why EWC improves performance even when Fisher estimates are purely local.

Limitations of local Fisher estimation. At $\alpha = 0.01$, each client receives very few classes, making its local FIM a poor estimate of the true task importance. For example, a client holding only class 0 and class 1 of MNIST will assign near-zero Fisher importance to weights in the output layer responsible for classes 2–9. When EWC is applied with this biased FIM, those weights are incorrectly treated as unimportant and overwritten by Task B gradients. Federated Fisher Aggregation (Section IV-E) addresses this by pooling FIM estimates across all clients, recovering a less biased global importance estimate.

TABLE II
SUMMARY OF EXPERIMENTAL RESULTS (SELECTED CONFIGURATIONS). BEST ACCURACY PER DATASET GROUP IN BOLD.

Method	Dataset	#Clients	Heterogeneity	#Rounds	Test Acc. (%)	BWT (%pts)	Conv. Round	Comm. Cost (MB)
FedAvg	MNIST	50	IID	120	65.61	−40.42	103	10949.9
FedAvg+EWC (local)	MNIST	50	IID	120	95.79	−1.60	107	10949.9
FedAvg+EWC	MNIST	100	IID	120	95.62	−2.22	108	21899.9
FedAvg	MNIST	10	$\alpha = 0.01$	120	47.85	−36.65	100	2190.0
FedAvg+EWC (local)	MNIST	10	$\alpha = 0.01$	120	73.57	−2.67	107	2190.0
FedAvg	FMNIST	50	IID	120	63.69	−40.27	106	10949.9
FedAvg+EWC (local)	FMNIST	50	IID	120	92.75	−3.32	108	10949.9
FedAvg+EWC	FMNIST	100	IID	120	94.03	−2.57	108	21899.9
FedAvg	FMNIST	10	$\alpha = 0.01$	120	44.99	−37.21	98	2190.0
FedAvg+EWC (local)	FMNIST	10	$\alpha = 0.01$	120	73.37	−2.12	106	2190.0
FedAvg	CIFAR-10	50	IID	120	54.02	−34.36	107	24554.3
FedAvg+EWC (local)	CIFAR-10	50	IID	120	78.27	−3.04	106	24554.3
FedAvg+EWC	CIFAR-10	100	IID	120	79.41	−2.14	108	49108.6
FedAvg	CIFAR-10	10	$\alpha = 0.01$	120	38.36	−30.73	100	4910.9
FedAvg+EWC (local)	CIFAR-10	10	$\alpha = 0.01$	120	69.53	−1.61	108	4910.9

Communication overhead. EWC adds no per-round communication overhead over FedAvg. The Fisher matrix is computed locally and stays on the client (or is aggregated once per task boundary in the federated variant). The per-task FIM aggregation costs one additional round carrying a vector of size $|\theta|$ floats, identical to a standard model update—negligible at 0.8% of total communication cost for a 3-task setup with 40 rounds per task.

Task construction and scalability. The permuted-task construction ensures equal task difficulty and zero class-overlap, making it a clean benchmark for forgetting. In real-world FL scenarios, tasks would correspond to genuinely different data distributions (e.g., different hospital protocols, different language dialects). Scaling to more tasks would require Online EWC [4] or a task-memory compression scheme, as the EWC penalty grows linearly with the number of completed tasks.

CIFAR-10 results. The small CNN has limited representational capacity for 3-task continual learning on 32×32 RGB images. Nonetheless, EWC reduces BWT from −34.4 to −3.0 points on CIFAR-10 (50 clients, IID)—a 31-point improvement even on this harder task, confirming that the Fisher regularization mechanism is not architecture-specific.

VIII. CONCLUSION

We presented a systematic study of Elastic Weight Consolidation in the federated continual learning setting, implemented using the Flower framework. Our modular codebase cleanly separates client-side Fisher consolidation (triggered by a server-sent `finalize_task` flag), task-aware server evaluation with per-task accuracy logging, and a Federated Fisher Aggregation extension that corrects non-IID bias in importance estimation. Across 90 experimental configurations, EWC consistently reduces catastrophic forgetting by approximately $10 \times$ compared to FedAvg, as measured by CFR and BWT. The proposed Federated Fisher Aggregation provides

further gains under high data heterogeneity ($\alpha = 0.01$) by pooling FIM estimates across clients.

Future work includes: (1) combining EWC with FedProx to jointly address client drift and catastrophic forgetting via a unified proximal-EWC objective; (2) exploring Online EWC [4] for settings with more than three tasks; (3) extending to natural language benchmarks (Sent140, Shakespeare) where task structure is more complex; and (4) studying adaptive λ schedules per client, motivated by APFL-style personalization [5].

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